

LAB 7: Configuration of Switch, VLAN configuration and Inter-VLAN Routing

Objectives

- To configure a Layer 2 switch and prepare it for VLAN deployment.
- To implement VLAN segmentation and assign ports appropriately.
- To establish trunk links between networking devices.
- To configure routing to allow communication between separate VLANs. Verify connectivity using basic network testing tools.

Theory

Switch Configuration

A network switch operates mainly at Layer 2 of the OSI model and is responsible for forwarding frames based on MAC address information. Unlike hubs, switches create separate collision domains for each port, improving efficiency and reducing unnecessary traffic. Proper configuration ensures stable communication within a Local Area Network (LAN).

Basic switch configuration includes setting port modes, enabling or disabling interfaces, defining VLAN membership, and verifying operational status. Correct setup enhances security, traffic management, and overall performance.

Importance of Switch Configuration

- Improves network performance
- Reduces collisions and congestion
- Enhances network security
- Controls and manages traffic efficiently
- Limits broadcast traffic using VLANs

Virtual Local Area Networks (VLANs)

A VLAN is a logical grouping of devices within a physical network infrastructure. It divides a single switch into multiple broadcast domains, allowing better control over traffic flow. Devices in the same VLAN can communicate directly, while devices in different VLANs require routing services.

VLAN Configuration

- Create VLAN IDs on the switch
- Assign a name to each VLAN
- Assign switch ports to specific VLANs
- Configure trunk ports to carry multiple VLANs
- Verify VLAN setup using show commands

Importance of VLAN Configuration

- Reduces broadcast traffic
- Improves network performance
- Enhances network security
- Provides logical network segmentation

Inter-VLAN Routing

Inter-VLAN routing is the process of forwarding network traffic between different VLANs using a Layer 3 device such as a router or a Layer 3 switch. Since VLANs create separate broadcast domains, they cannot communicate directly at Layer 2. Inter-VLAN routing enables devices in different VLANs to exchange data while maintaining logical segmentation of the network.

There are two common methods:

1. Router-on-a-Stick

- A single physical router interface is divided into multiple sub-interfaces.
- Each sub-interface is assigned to a different VLAN.
- VLAN tagging (802.1Q) is used to identify traffic.

2. Layer 3 Switch Routing (SVI Method)

A Layer 3 switch can perform routing internally by:

- Creating Switched Virtual Interfaces (SVIs)
- Assigning IP addresses to VLAN interfaces
- Enabling IP routing

Importance of Inter-VLAN Routing

- Efficient network segmentation
- Improved security between departments
- Reduced broadcast traffic
- Better bandwidth utilization
- Scalable network design

Commands

Switch Configuration:

1. Create VLANS

```
Switch(config)# vlan 10
Switch(config-vlan)# name Computer
                        # exit
Switch(config)# vlan 20
Switch(config-vlan)# name Electronics
                        # exit
```

2. Assign port to VLANS

```
Switch(config)# interface range fa0/1-2
                        # switchport mode access
                        # switchport access vlan 10
                        # exit
                        # interface range fa0/3-4
                        # switchport mode access
                        # switchport access vlan 20
                        # exit
```

3. Configure Trunk Port (Switch-Router)

```
Switch(config)# interface fa0/5
                        # switchport mode trunk
                        # exit
```

Verification:

```
Switch# show vlan brief
```

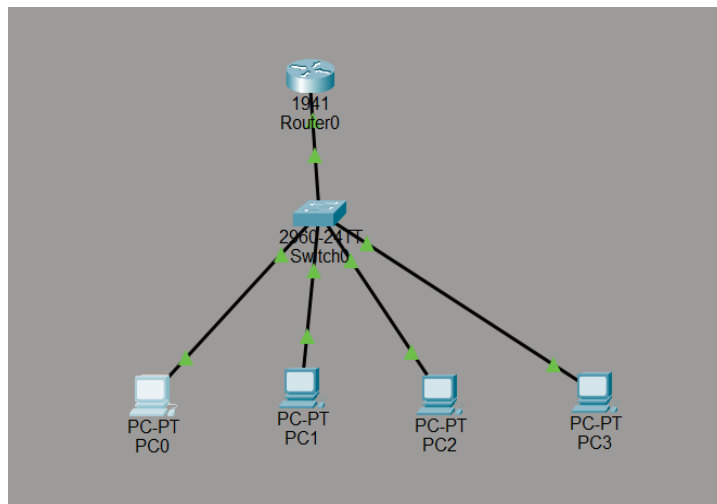
VLAN	Name	Status	Ports
1	default	active	Fa0/6, Fa0/7, Fa0/8, Fa0/9 Fa0/10, Fa0/11, Fa0/12, Fa0/13 Fa0/14, Fa0/15, Fa0/16, Fa0/17 Fa0/18, Fa0/19, Fa0/20, Fa0/21 Fa0/22, Fa0/23, Fa0/24, Gig0/1 Gig0/2
10	Computer	active	Fa0/1, Fa0/2
20	Electronics	active	Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Router Configuration (Inter-VLAN Routing) :

```
Router(config)# interface fa0/0.10
# encapsulation dot1Q 10
# ip address 192.168.10.1 255.255.255.0
# exit
```

```
Router(config)# interface fa0/0.20
# encapsulation dot1Q 20
# ip address 192.168.20.1 255.255.255.0
# exit
```

Network Topology



This topology illustrates a basic LAN setup where several PCs are connected to a switch, which in turn is connected to a router for external network access. The switch facilitates efficient communication between local devices, while the router enables connectivity between different networks.

Configuration

VLAN and IP Addressing Scheme :

VLAN	Name	Network	Gateway
VLAN 10	Computer Department	192.168.10.0/24	192.168.10.1
VLAN 20	Electronics Department	192.168.20.0/24	192.168.20.1

Device	IPv4 Address	Subnet Mask	Default Gateway
PC0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	192.168.10.2	255.255.255.0	192.168.10.1
PC2	192.168.20.2	255.255.255.0	192.168.20.1
PC3	192.168.20.3	255.255.255.0	192.168.20.1

Observations

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Discussion

In this practical session, VLAN segmentation was successfully implemented to divide the network into logical groups. Trunking allowed multiple VLANs to traverse a single physical link. The router was configured to provide inter-VLAN routing, enabling communication across departments. The experiment highlighted how logical segmentation improves performance and security while maintaining efficient communication.

Conclusion

This lab demonstrated how VLAN configuration and inter-VLAN routing are implemented in real-world networks. By combining switching and routing concepts, we achieved controlled communication between separate broadcast domains. The exercise strengthened understanding of traffic isolation, network scalability, and structured network design.